MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MICROWAVE

TYPE 6173

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Diode, pencil, glass and metal
See figure 1
Mounting position: Any
Weight: 0.2 ounce (5.67 grams) nominal

ABSOLUTE RATINGS:

Parameter: / Unit:	F MHz	Ef V	epx v	epy v	ik a	Ib mAdc (Note 1)	(E) ib a	Io mAdc
Half-wave rectifier:								
Maximum:	---	6.9	375	---	0.05	---	---	5.5
Minimum:	---	5.7	---	---	---	---	---	---
Pulse detector:								
Maximum:	3,300	6.9	1,000	150	1.0	1.0	Note 1	---
Minimum:	---	5.7	---	---	---	---	---	---
TEST CONDITIONS:	---	6.3	---	---	---	---	---	---

ABSOLUTE RATINGS:

Parameter: / Unit:	Ehk Vdc	T(seal) °C	tk sec	Epp Vac	Barometric pressure, reduced mmHg
Half-wave rectifier:					
Maximum:	±90	175	---	---	54.24
Minimum:	---	---	60	---	---
Pulse detector:					
Maximum:	±90	175	---	---	54.24
Minimum:	---	---	60	---	---
TEST CONDITIONS:	---	---	180	117	---

GENERAL:

(E) Qualification - Not required

Holding period - t = 96 hours

(E) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1	2							
1201	Tap shorts	3		1.0	II	---	---	---	---
1231	Emission	-	Rk = 0; Eb = 5 V	0.65	II	Is	11	---	mAdc
1231	Pulse emission	4	Rk = 0; eb = 100 V (max); tp = 1.0 ±0.1 µs; prr = 1,000	0.65	II	ib	0.5	1.3	a
1331	Direct-interelectrode capacitance	-	Test in jig in accordance with Drawing 155-JAN	0.65	II	Cpk	0.8	1.4	pF
	Quality conformance inspection, part 2								
1301	Heater current	-		---	---	If	125	145	mA
1336	Heater-cathode leakage	-		---	---	Ihk	---	±100	µAdc
1211	Insulation of electrodes	-	Eb = -300 V	---	---	Rpk	25	---	Meg
(E) ---	Operation, pulse detector	5		---	---	etd	50	---	mv
	Quality conformance inspection, part 3								
---	Life test	6	Group B; Ehk = 100 V; t = 500 hours	---	---	---	---	---	---
---	Life-test end point:								
---	Operation of rectifiers	6		---	---	Io	3.5	---	mAdc
---	Operation of rectifiers	6,11	Ehk = 0	---	---	Io	3.7	---	mAdc
1031	High-frequency vibration	7,11	No voltages; F = 40 to 60 Hz; 10 G; t = 180 sec	---	---	---	---	---	---
1002	Barometric pressure, reduced	8,11	Pressure = 55 ±5 mmHg; voltage = 300	---	---	---	---	---	---
---	Torque	9,11	No voltages	---	---	---	12.5	---	Lb/in.
1116	Lead fatigue	11	No voltages; each heater lead	---	---	---	4	---	arcs
---	RF loading resistance	10,11	No voltages	---	---	R	100,000	---	Ohms

NOTES:

1. For operation at maximum pulse rating of ib = 1.0 amp, tp shall not exceed 1.5 µs and prr shall not exceed 700 pulses per second. Higher values of tp and prr are permissible provided the average anode current (Ib) does not exceed 1.0 mAdc.
2. All tests listed in quality conformance inspection, part 1, are to be performed after the holding period.
3. Tubes shall be tested in accordance with method 1201, except that the tube shall be mounted on a paddle and the paddle tapped with a rubber tapper three times in three mutually perpendicular axes.
4. In measuring or recording the applied pulse voltage (eb), and the peak-pulse current (ib), the smooth peak value shall apply. The smooth peak is the maximum value of a smooth curve through the average of the fluctuations over the top portion of the pulse. Spike voltages shall be excluded.

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NOTES: -Continued

5. Test in diode detector cavity per Drawing 228-JAN. Cavity bandwidth to be 50 MHz minimum. Conditions: Rk = 1,500 ohms; tp = 10 µs; prr = 1,000; pi = 1 mw, 100 percent modulated into 50-ohm load; F = 3,300 ±100 MHz; Ef = 6.3 V.

6. Use half-wave rectifier circuit. Frequency = 60 Hz. Series impedance (Zp) including the transformer shall be approximately 300 ohms. Load impedance = 30,000 ohms, load capacitance = 8 µF.

7. Tubes shall be vibrated in a plane perpendicular to the tube axis only. At the end of the vibration period the tubes shall meet the specified limits for If, Ihk, and Is.

8. The specified voltage shall be applied between anode and cathode, F = 60 Hz; Ef = 0. Tubes showing evidence of corona or arcing shall be rejected as inoperable. This is a nondestructive test.

9. The cathode terminal shall be rigidly clamped and a torque of 12.5 lb-in. shall be applied gradually about the tube axis to the anode terminal. The tube shall not fracture. This is a destructive test. Electrical rejects may be used.

10. The rf cold-loading resistance of the tube is measured in the following manner: Ef = 0; a 450 MHz coaxial cavity, RCA Model H1821B, or equivalent, shunted by a calibration resistor of 97 K ohms is employed. The relative Q of this circuit is determined by applying a modulated rf signal into the cavity. The tube is then inserted into the cavity, parallel with the calibration resistor. Using the resistance-variation method, the rf loading resistance of the tube is calculated from the following formula:

$$Rx = (97K)\left[\log\left(\frac{\Delta A}{20} - 1\right)\right]$$

where Rx = rf loading resistance of the tube (K ohms)
ΔA = change in relative Q of the cavity (dB)

(E) 11. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, MU
Air Force - 11, 17, 80
DSA - ES

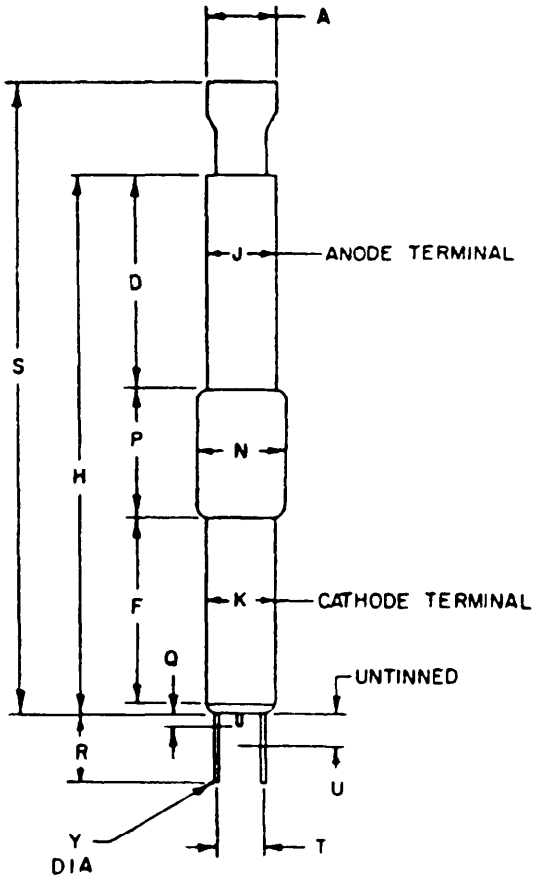
User activities:
Navy - AS, OS, MC, CG, SH
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Preparing activity:
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 2			
A		.230 (5.84)	
D	.580 (14.73)		
F	.580 (14.73)		
H	1.590 (40.39)	1.675 (42.55)	
N		.320 (8.13)	
P		.415 (10.54)	
Q		.040 (1.02)	
R	.200 (5.08)	.240 (6.10)	
S		1.987 (50.47)	
T	.075 (1.91)	.155 (3.94)	
Y	.018 (.46)	.022 (.56)	
Quality conformance inspection, part 3			b (E)
J	.245 (6.22)	.255 (6.48)	a, e
K	.245 (6.22)	.255 (6.48)	d
U		.080 (2.03)	c
External surfaces			f

NOTES:

a. Maximum allowable eccentricity of the anode terminal with respect to the cathode terminal shall be .010 (.25 mm). Eccentricity of the anode shall be gaged by chucking the cathode terminal .050 (1.27 mm) minimum and .100 (2.54 mm) maximum from the cathode-glass seal and gaging 1/2 TIR, .080 (2.03 mm) from the end of the anode terminal.

b. These dimensions shall be inspected every three months during production. At start of production, a sample of six tubes selected at random shall meet acceptance requirements with no allowable defects. Failure of any tube dimension to meet acceptance requirements shall cause that dimension to be inspected to an AQL of 2.5 percent individual and 4.0 percent combined. When three successive samples as specified above meet acceptance requirements for this dimension, the dimension shall revert to the 3 month sampling plan contained herein.

c. Heater leads shall be trimmed, and shall be tinned except as specified by dimension U. Untinned lengths of leads shall be measured from end of cathode terminal.

d. Glass from stem sealing shall not be present on cathode terminal surface. Glass button shall not extend radially beyond cathode terminal surface.

e. Diameter of anode terminal shall not be increased by solder or melted plating from exhaust tubulation.

f. All external metal surfaces, except leads and exhaust tubulation, shall be silver or gold plated 50 msi minimum. Plating shall not be blistered and plated areas shall exhibit no exposed metal.

FIGURE 1. Outline drawing of electron tube type 6173.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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